**HWA CHONG INSTITUTION**
**JC2 Preliminary Examinations**
**Higher 3**

# MATHEMATICS

**9824/01**

Paper 1

**18 September 2015**

**3 hours**

Additional Materials: Answer Paper
List of Formulae (MF15)

## READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **6** printed pages.

**1** Let $\{x_n\}$ be a sequence of real numbers where

$$x_{n+2} = \frac{1}{3}x_{n+1} + \frac{2}{3}x_n \text{ for } n \in \mathbb{N} \cup \{0\} \text{ and}$$

$$x_n = \sum_{r=0}^{n-1}\left(-\frac{2}{3}\right)^r + \frac{4}{3}\sum_{r=0}^{n-2}\left(-\frac{2}{3}\right)^r \text{ for } n \geq 2, n \in \mathbb{Z}^+ \text{--------------- } (*)$$

It's given that $x_0 = 2$ and $x_1 = 1$.

**(i)** Show that (*) is true for $n = 2$ and $n = 3$. [2]

**(ii)** Prove by induction that (*) is true for $n \geq 2, n \in \mathbb{Z}^+$. [6]

**(iii)** Evaluate $\lim_{n\to\infty} x_n$. [2]

**2** **(a)** Let $\mathrm{f} : \mathbb{R}\setminus\{0\} \to \mathbb{R}\setminus\{0\}$ be a strictly decreasing bijective function.

**(i)** Show that $\mathrm{f}^{-1}$ is also a strictly decreasing function. [2]

**(ii)** Let $n \in \mathbb{Z}^+$. Prove that if $a_1 \leq a_2 \leq \ldots \leq a_n$, then

$$\mathrm{f}^{-1}(a_n) \leq \frac{1}{n}\sum_{k=1}^{n}\mathrm{f}^{-1}(a_k) \leq \mathrm{f}^{-1}(a_1).$$ [3]

**(iii)** Let $\mathrm{g} : \mathbb{R}\setminus\{0\} \to \mathbb{R}\setminus\{0\}$ be defined by $\mathrm{g}(x) = \dfrac{1}{x^{\frac{1}{3}}}$. Show that g is bijective and strictly decreasing. By finding $\mathrm{g}^{-1}(x)$ and using the result in part **(ii)**, show that

$$\frac{1}{n^3} \leq \frac{1}{n}\left(1 + \frac{1}{2^3} + \frac{1}{3^3} + \ldots + \frac{1}{n^3}\right) \leq 1 \text{ for } n \in \mathbb{Z}^+.$$ [5]

**(b)** **(i)** Evaluate $\displaystyle\sum_{r=1}^{n}\frac{1}{r(r+1)}$. [2]

**(ii)** Hence, by consider $\dfrac{1}{1\cdot 2}, \dfrac{1}{2\cdot 3}, \ldots$ and using Arithmetic-geometric mean inequality, prove that

$$n! \geq (n+1)^{\frac{n-1}{2}} \quad \forall n \in \mathbb{Z}^+.$$ [3]

**3** **(i)** Show that $\frac{\pi}{2}-\tan^{-1}\left(\sqrt{2}+1\right)=\tan^{-1}\left(\sqrt{2}-1\right)$. [2]

**(ii)** By using $t^4+1=\left(t^2-\sqrt{2}t+1\right)\left(t^2+\sqrt{2}t+1\right)$ and part **(i)**, show that

$$\int_0^1 \frac{1}{1+t^4}\,\mathrm{d}t=\frac{1}{4\sqrt{2}}\left[\ln\left(2+\sqrt{2}\right)-\ln\left(2-\sqrt{2}\right)+\pi\right].$$ [8]

**(iii)** It is given that $0\le x\le 1$.

**(a)** By using the fact that $\frac{1}{1+t^4}\le 1$ for all $t\ge 0$, show that $\int_0^x \frac{t^{4n}}{1+t^4}\,\mathrm{d}t\le\frac{x^{4n+1}}{4n+1}$. [2]

**(b)** Hence show that $\lim_{n\to\infty}\int_0^x \frac{t^{4n}}{1+t^4}\,\mathrm{d}t=0$. [3]

**4** Codewords of length $m$ are generated from the digits {0, 1, 2, 3}.

**(i)** If the order of the digits within each codeword does not matter,

**(a)** how many different codewords are there? [2]

**(b)** how many different codewords with equal number of even and odd digits are there? [3]

**(ii)** Assuming that the order of the digits within each codeword matters, a codeword is considered as legitimate if and only if it has an even number of 0s. Find a recurrence relation for $a_n$, where $a_n$ is the number of legitimate codewords of length $n$. Hence find $a_4$. [4]

Suppose now codewords of length $m$ are made from the digits {1, 2, 3, … , $p$}, and the order of the digits within the codeword matters.

**(iii)** How many different codewords are there without using the digits from {1, 2, 3, …, $q$ }, where $q<p$ ? [1]

**(iv)** By using the principle of inclusion and exclusion, prove the binomial theorem

$$(p-q)^m=\sum_{r=0}^{m}\binom{m}{r}(-1)^r q^r p^{m-r}$$ [3]

**5** Consider an equilateral triangle of side length *n*, which is divided into rows of unit triangles. A path travelling from the top triangle to the middle triangle in the last row must satisfy the following two conditions:

(1) The path must travel from one triangle to next triangle which shares a common edge.
(2) The path never travels to the row above or revisit a triangle.

Figure 1 shows an example of a path for a triangle of side length 5.

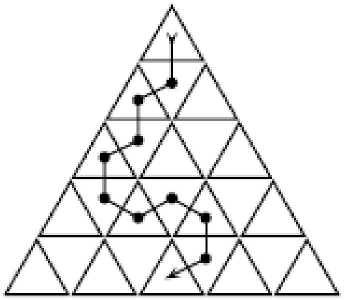
Figure 1

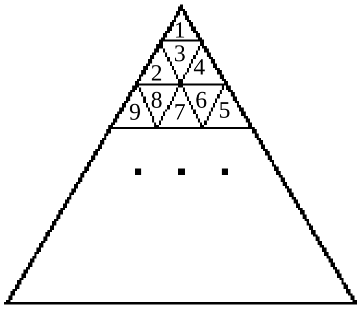

Figure 2

**(i)** By using the Bijection Principle, show that the total number of different paths $f(n)=(n-1)!$ [2]

**(ii)** The length of a path is defined as the number of triangles in that path.

**(a)** Find the shortest length of all possible paths and show that the longest path has a length of $n^2-2n+2$. [3]

**(b)** For $n=8$, show that there exists at least one number *m*, where the number of paths with length *m* exceeds 136. [2]

**(iii)** A colouring is defined to be the instance when *r* unit triangles, where $r \geq 1$, are coloured in red and the rest of the unit triangles are not coloured.

**(a)** How many different colourings are there that do not form a path? [2]

**(b)** The unit triangles are labelled in ascending order as shown in Figure 2. For $n=8$, find the number of paths that has at most one perfect square greater than 1, e.g. 4, 9, 16 etc. [3]

**6** A particular solution of the differential equation

$$\frac{dy}{dx} = x + y + 1$$

has $y = 2$ when $x = 1$.

**(i)** Use the Euler's method with step size $h = 0.2$ to estimate $y$ at $x = 1.6$. State with a reason whether this value of $y$ is an under-estimate or an over-estimate. [4]

**(ii)** Using the improved Euler's method with step size $0.2$, the approximate value of $y$ at $x = 1.2$ is $2.9$. Carry out two more applications of the improved Euler's method to estimate $y$ at $x = 1.6$. [3]

**(iii)** The general solution of the differential equation is $y = Ae^{x} - (x+2)$, where $A$ is a constant. Find the value of $y$ at $x = 1.6$. [2]

**(iv)** Use the results in the earlier parts to show that the improved Euler's method gives a better approximation of $y$ at $x = 1.6$ than the Euler's method for this differential equation. [1]

**7** An accident has caused $7000$ kg of potassium permanganate to spill into a mountain lake, colouring its $10^5$ litres of water purple. The chemical also contaminates a nearby stream which feeds the lake at 900 litres per minute of water containing $0.12$ kg of potassium permanganate per litre. Another stream takes water from the lake at the rate of 1000 litres per minute.

Find the concentration of potassium permanganate in the lake as a function of time. [6]

**8** A system consisting of a spring in a viscous fluid is installed so as to absorb the shock on a 100kg mass. The second order differential equation that describes this scenario is given by

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 25x = 0,$$

where $x$ is the displacement (in cm), as a function of time $t$ (in second), of the mass relative to its equilibrium position.

**(i)** Show that the system is underdamped. [1]

Suppose a shock is sustained when the system is in equilibrium imparting an instantaneous velocity of 200 cm/s.

**(ii)** Find the exact time (in second) at which the amplitude of the oscillation first falls below 1 cm. [6]

9 The common toads (Bufo bufo) compete with each other for food and space and the negative effects of competition depend on how often toads meet each other. It is known that Bufo Bufo is an important food source for the grass snake (Natrix natrix).

**(a)** In the absence of grass snakes, the differential equation for modelling the population dynamics of Bufo bufo is given by

$$\frac{\mathrm{d}B}{\mathrm{d}t} = rB\left(1-\frac{B}{K}\right).$$

where $r$ and $K$ are constants, and $B$ is the number of Bufo bufo per hectare of land in $t$ months.

**(i)** Explain, in context, the meaning of the two constants in the differential equation. [2]

**(ii)** Find the number of Bufo bufo per hectare of land at the instant when the population has the greatest increment. What happens to the population after this instant? Leave your answers in terms of $K$. [3]

**(b)** In the presence of grass snakes, the population of Bufo bufo per hectare of land in $t$ months is modelled by

$$\frac{\mathrm{d}B}{\mathrm{d}t} = rB\left(1-\frac{B}{K}\right)-\frac{1}{r}.$$

**(i)** Explain, in context, the meaning of $\frac{1}{r}$. [1]

Take $K = 100$.

**(ii)** Sketch a bifurcation diagram for the parameter $r$ and find the bifurcation value of $r$. [5]

**(iii)** State the range of values of $r$ such that the population of Bufo bufo will become depopulated regardless of the initial population. [1]

**[End of Paper]**